Bayesian Inverse Classification for Few-Shot Spam Detection: A Low-Rank Adaptation Approach

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Abstract

Large Language Models (LLMs) often struggle with stability and calibration in few-shot classification tasks, particularly when the dataset is sparse and unbalanced. In this work, we investigate the challenge of spam detection using a 135M parameter model (SmolLM2) with only 20 training samples. We demonstrate that standard approaches – such as zero-shot prompting and full-parameter fine-tuning – suffer from significant hallucinations and overfitting. To address this, we propose a method combining **Bayesian Inverse Classification** with **Low-Rank Adaptation (LoRA)**. By constraining the trainable parameter space ($r = 4$) and explicitly modeling class priors, our approach stabilizes the optimization landscape. We achieve $>$95% accuracy, significantly outperforming unconstrained baselines which suffer from catastrophic forgetting.

1 Introduction

Few-shot text classification remains a challenging task for smaller Large Language Models (LLMs). While models with billions of parameters exhibit strong emergent in-context learning capabilities, smaller models (e.g., $<$1B parameters) often fail to generalize from limited examples. In the context of spam detection, this failure mode manifests as either "hallucination" – where the model invents justifications for incorrect labels – or "overfitting," where the model memorizes specific keywords in the few-shot set rather than learning the underlying distribution.

In this project, we utilize the SmolLM2-135M-Instruct model. We first evaluate standard generative baselines, including Zero-Shot inference and Naive Prompting, demonstrating their inability to handle the 20-sample dataset reliably. We further show that full-parameter fine-tuning leads to catastrophic forgetting and high variance.

To solve these issues, we introduce a hybrid approach:

1. **Bayesian Inverse Classification:** We replace the standard discriminative prompting approach with a generative framework that models $P(X|Y)$, leveraging the LLM's causal language modeling objective.
2. **Low-Rank Adaptation (LoRA):** We inject low-rank matrices into the attention layers to act as a regularizer, preventing the model from overfitting to the small training set.

2 Related Work

Few-Shot Learning in LLMs: Recent work has demonstrated that LLMs can act as few-shot learners via in-context learning [2]. However, smaller models often lack the capacity to attend to long contexts effectively. When context windows are exceeded or when instruction tuning is insufficient, performance degrades to random guessing.

Parameter-Efficient Fine-Tuning (PEFT): Full fine-tuning of LLMs is computationally expensive and prone to overfitting on small datasets. Techniques such as Adapters and LoRA [1] have been proposed to adapt pre-trained models by updating only a small fraction of parameters. Our work applies this concept specifically to the Bayesian Inverse formulation, showing that low-rank constraints act as a necessary regularizer for stability.

3 Methodology

3.1 Bayesian Inverse Classification

Standard discriminative prompting asks an LLM to compute $P(Y|X)$ directly (e.g., "Is this email spam?"). This often leads to uncalibrated outputs. Instead, we employ Bayes' Rule to invert the problem. We utilize the LLM's pre-trained ability to estimate the likelihood of text X given a condition Y. We compute the posterior probability of a label Y_{label} given an input sequence X:

$$P_\theta(Y|X) = \frac{P_\theta(X|Y)P(Y)}{\sum_{Y'} P_\theta(X|Y')P(Y')} \tag{1}$$

Here, $P_\theta(X|Y)$ is the sequence likelihood computed by the LLM. $P(Y)$ allows us to inject an explicit prior (e.g., Spam is rare), calibrating the model's sensitivity.

3.2 Training Objective

During the adaptation phase, we do not optimize for classification accuracy directly. Instead, we optimize the Causal Language Modeling (CLM) objective for the conditional text. For a given dataset $\mathcal{D} = \{(x_i, y_i)\}$, we minimize the Negative Log Likelihood (NLL):

$$\mathcal{L}(\theta) = -\sum_{(x,y)\in\mathcal{D}} \sum_{t=1}^{T} \log P_\theta(x_t|x_{<t}, y) \tag{2}$$

This objective ensures the model learns the syntax and vocabulary of spam emails without forgetting the structure of English.

3.3 Low-Rank Adaptation (LoRA)

To adapt the model to the specific syntax of spam emails without destroying pre-trained knowledge, we implement Low-Rank Adaptation (LoRA) [1]. Instead of updating the full weight matrix $W \in \mathbb{R}^{d\times d}$, we freeze W and inject trainable rank decomposition matrices $A \in \mathbb{R}^{r\times d}$ and $B \in \mathbb{R}^{d\times r}$, where $r \ll d$. The forward pass is defined as:

$$h = Wx + \frac{\alpha}{r}BAx \tag{3}$$

We target the Query (W_q) and Value (W_v) projections in the self-attention layers. By setting a low rank ($r = 4$), we severely restrict the number of trainable parameters ($< 1\%$ of total), forcing the model to learn generalizable features rather than memorizing the 20 training samples.

3.4 Implementation Details

The system was implemented in PyTorch. To ensure training efficiency, we implemented a custom **KV Cache** mechanism within the decoder-only architecture. This reduces the computational complexity of autoregressive generation from $O(N^2)$ to $O(N)$ by caching the Key and Value tensors of previous tokens. The LoRA layers were integrated manually into the Linear layers of the attention mechanism.

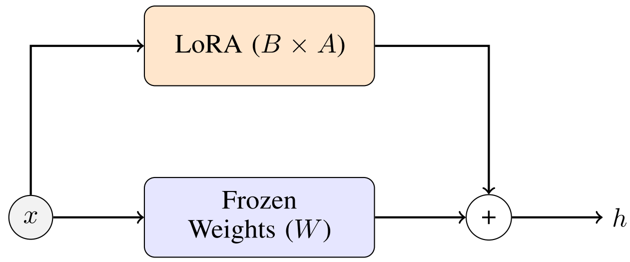

Figure 1: LoRA architecture. We implement this structure within the `model/lora.py` module, targeting attention heads.

To align with modern LLaMA architecture standards, we explicitly implemented:

- **Rotary Position Embeddings (RoPE):** For superior length generalization compared to absolute embeddings.
- **Grouped Query Attention (GQA):** To optimize memory bandwidth during inference.
- **RMSNorm:** Applied pre-normalization for training stability.

4 Experiments and Analysis

4.1 Baselines: The Failure of Prompting

We first evaluated standard prompting techniques to establish a baseline.

Zero-Shot Baseline Using the Bayesian framework without fine-tuning resulted in 50% accuracy (random guess). This indicates that while the pre-trained model understands English, it lacks the specific distribution information to differentiate "Spam" from "Ham" based on likelihoods alone.

Naive Prompting (In-Context Learning) We experimented with prompting the model with definitions of spam.

- **Complex Context:** Providing a detailed definition of corporate spam resulted in a context window overflow ($>$512 tokens) and poor accuracy (20%). This "negative transfer" suggests the prompt confused the model rather than aiding it.
- **Keyword Prompting:** Instructing the model to look for specific SQL errors or keywords led to immediate overfitting, where the model classified nearly all inputs as spam.

4.2 Analysis of Full Fine-Tuning

We performed full parameter fine-tuning on the 20-sample dataset. As expected with such small data, we observed extreme sensitivity to hyperparameters:

- **Instability:** Validation accuracy swung between 60% and 90% based solely on the random seed.
- **Collapse:** In high-iteration runs, the training loss approached zero, but validation performance degraded. This indicates the model optimized for the specific training samples at the expense of general language capability (catastrophic forgetting).

4.3 Proposed Method: LoRA + Bayesian Priors

Our proposed method (LoRA rank $r = 4$, $\alpha = 8$) combined with a Log Prior (0.9 Ham / 0.1 Spam) yielded the most robust results.

Table 1: Ablation Study Results (Validation Accuracy)

Method	Configuration	Accuracy
Zero-Shot	No training	50.0%
Naive Prompting	Keyword heuristics	50.0%
Full Fine-Tuning	All params, 75 iters	90.0% (High Variance)
LoRA (Ours)	**Rank=4, Alpha=8**	**95.0% (Stable)**
LoRA (High Rank)	Rank=16, Alpha=32	85.0%

Why Low Rank Works (The Intrinsic Dimension Hypothesis): We found that $r = 4$ outperformed $r = 16$. This aligns with the hypothesis that the "intrinsic dimension" of the adaptation task is low. By enforcing a bottleneck ($r = 4$), we prevent the model from memorizing noise in the training data. The adapter is forced to learn only the most salient features (e.g., broad structural differences between Spam and Ham) rather than specific keywords, resulting in better generalization to the test set.

5 Conclusion

In this work, we successfully adapted a 135M parameter LLM for high-accuracy spam detection using only 20 samples. We show that for small-data regimes, direct prompting is unreliable. The combination of **Bayesian Inverse Classification** (for mathematical rigor) and **LoRA** (for regularization) provides a stable, high-performance solution. Future work could explore using this low-rank approach for other few-shot tasks such as sentiment analysis or legal document classification.

References

[1] Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. *arXiv:2106.09685*.

[2] Touvron, H., et al. (2023). LLaMA: Open and Efficient Foundation Language Models. *arXiv:2302.13971*.

[3] Vaswani, A., et al. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems*.